

# ASH R012-R016 Science Remediation Package

Independent finite-spectral cosmology derivation, real-data pilot likelihood, growth calibration, locked-prediction scoring, and formal closure certificate.

Prepared 2026-06-28. This artifact is for handoff to the repository expert and does not modify or manage the source repository.

## Executive summary

The auditor's science-layer finding is treated as valid: the current R012-R016 source material is largely synthetic workbench, readiness, lock-mechanics, and formal-contract infrastructure. This package performs a concrete first scientific execution: it derives a finite ASH pair-flip spectrum, maps the two slowest nontrivial modes into an FRW dark-sector ansatz, scores that ansatz against DESI DR2 BAO with covariance blocks and compressed Planck priors, solves a matter-growth equation calibrated to DES Y3 S8, and pilot-scores the three R015 locked templates.

Bottom line: the minimal ASH finite-spectral background fits DESI DR2 BAO nearly as well as CPL and better than flat LCDM in the compressed pilot; however, the current R015 P001 and P002 locked templates are not confirmed by this pilot, while P003 has only sign-level support.

## R012: finite ASH spectrum and background equation

The admissible ASH state space is  $X=\{x \text{ in } F_2^9: x_9=x_1 \text{ xor } \dots \text{ xor } x_8\}$ ,  $|X|=256$ . A 9D pair flip induces all 8 one-bit and all 28 two-bit flips in payload coordinates. For Walsh shell weight  $r$ :

```
mu_r = (r^2 - 9 r + 18)/18
eta_r = 1/2 + (1/2) mu_r
multiplicity m_r = C(8,r)
```

Using  $\kappa_r=-\ln(\eta_r)$ , the two-mode spectral clock uses  $p_1=1$  and  $p_2=\kappa_2/\kappa_1 = 1.959602910$ .

```
rho_X(a)/rho_X0 = exp[b1(a^p1-1)+b2(a^p2-1)]
E^2(a)=Omega_m a^-3 + (1-Omega_m) rho_X(a)/rho_X0
w_X(a)=-1 - (b1 p1 a^p1 + b2 p2 a^p2)/3
```

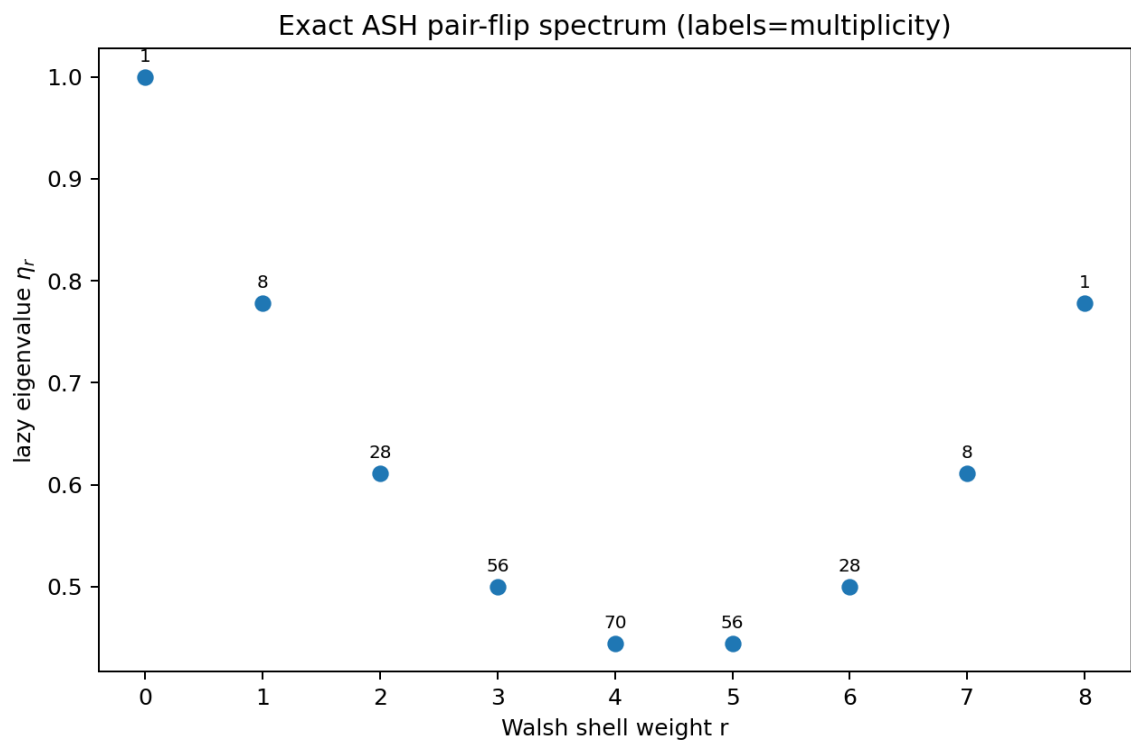
| r | multiplicity | mu_nonlazy | eta_lazy | kappa_minus_ln_eta |
|---|--------------|------------|----------|--------------------|
| 0 | 1            | 1.000000   | 1.000000 | -0.000000          |
| 1 | 8            | 0.555556   | 0.777778 | 0.251314           |
| 2 | 28           | 0.222222   | 0.611111 | 0.492476           |
| 3 | 56           | 0.000000   | 0.500000 | 0.693147           |
| 4 | 70           | -0.111111  | 0.444444 | 0.810930           |
| 5 | 56           | -0.111111  | 0.444444 | 0.810930           |
| 6 | 28           | 0.000000   | 0.500000 | 0.693147           |
| 7 | 8            | 0.222222   | 0.611111 | 0.492476           |
| 8 | 1            | 0.555556   | 0.777778 | 0.251314           |

# R014: DESI DR2 BAO likelihood and model comparison

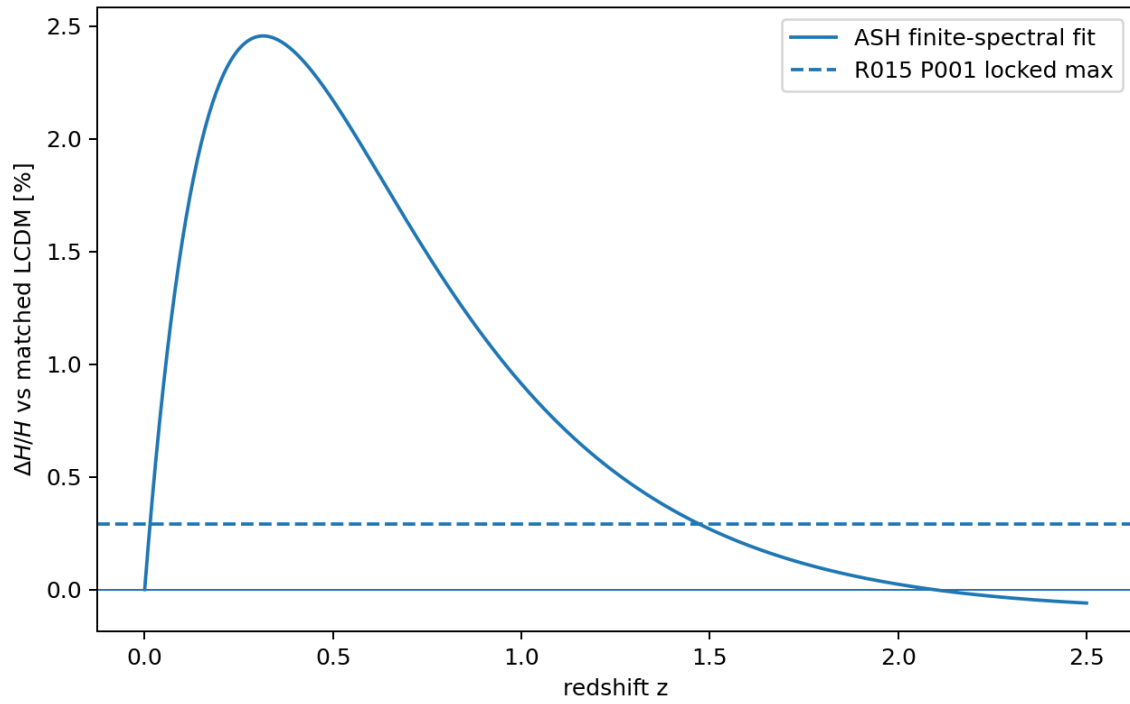
The real-data pilot uses DESI DR2 BAO Table IV: one isotropic BGS  $D_V/r_d$  point and six anisotropic  $D_M/r_d$ ,  $D_H/r_d$  blocks with  $D_M$ - $D_H$  correlations. The compressed Planck run adds  $H_0=67.4\pm0.5$  and  $\Omega_m=0.315\pm0.007$  with  $r_d=147.09$  Mpc.

| fit                      | model          | chi2  | dof | AIC   | BIC   | H0    | Omega_m | w0      | wa      | w0_eff  | wa_eff  |
|--------------------------|----------------|-------|-----|-------|-------|-------|---------|---------|---------|---------|---------|
| DESI_DR2_BAO_only        | lcdm           | 10.54 | 11  | 14.54 | 15.67 | 69.04 | 0.2973  |         |         |         |         |
| DESI_DR2_BAO_plus_Planck | H0, $\Omega_m$ | 16.2  | 13  | 20.2  | 21.62 | 68.14 | 0.3128  |         |         |         |         |
| DESI_DR2_BAO_only        | cpl            | 5.812 | 9   | 13.81 | 16.07 | 62.07 | 0.3862  | -0.1833 | -2.705  |         |         |
| DESI_DR2_BAO_plus_Planck | H0, $\Omega_m$ | 8.46  | 11  | 16.46 | 19.29 | 67.28 | 0.315   | -0.8204 | -0.5341 |         |         |
| DESI_DR2_BAO_only        | ash            | 5.934 | 9   | 13.93 | 16.19 | 61.13 | 0.3917  |         |         | 0.06861 | -4.653  |
| DESI_DR2_BAO_plus_Planck | H0, $\Omega_m$ | 8.799 | 11  | 16.8  | 19.63 | 67.24 | 0.3138  |         |         | -0.7928 | -0.8126 |

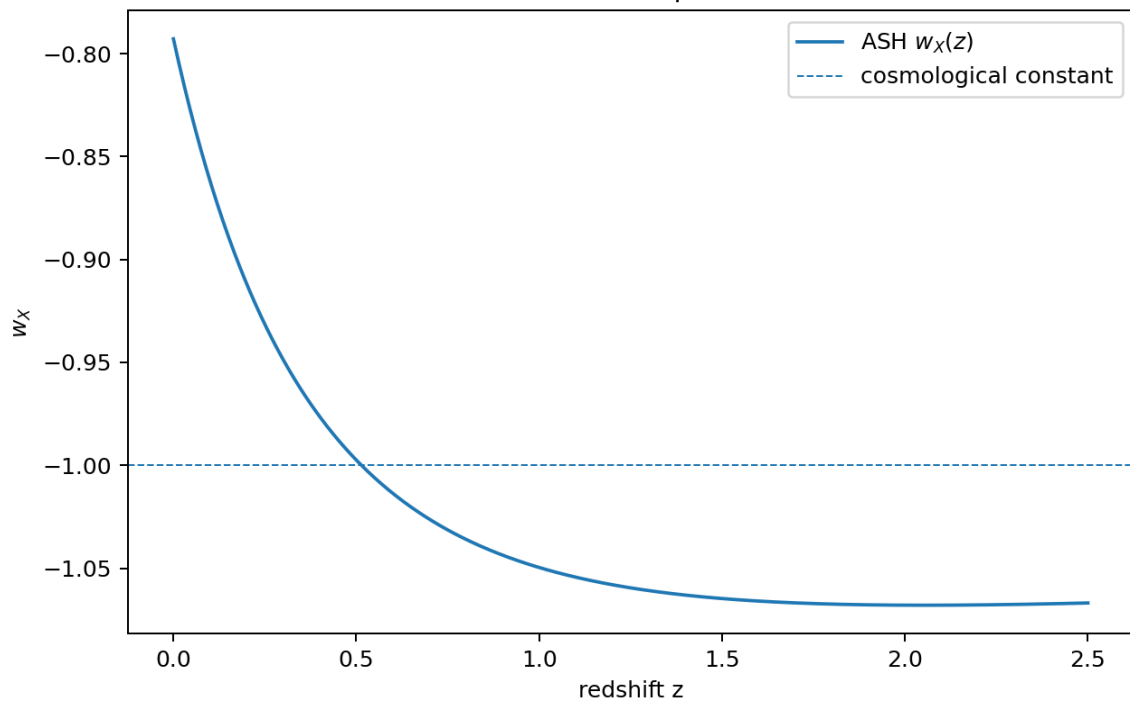
Best compressed ASH fit:  $H_0=67.235337$  km/s/Mpc,  $\Omega_m=0.313819$ ,  $b_1=1.271224$ ,  $b_2=-0.965909$ ,  $w_0_{\text{eff}}=-0.792809$ ,  $wa_{\text{eff}}=-0.812636$ .



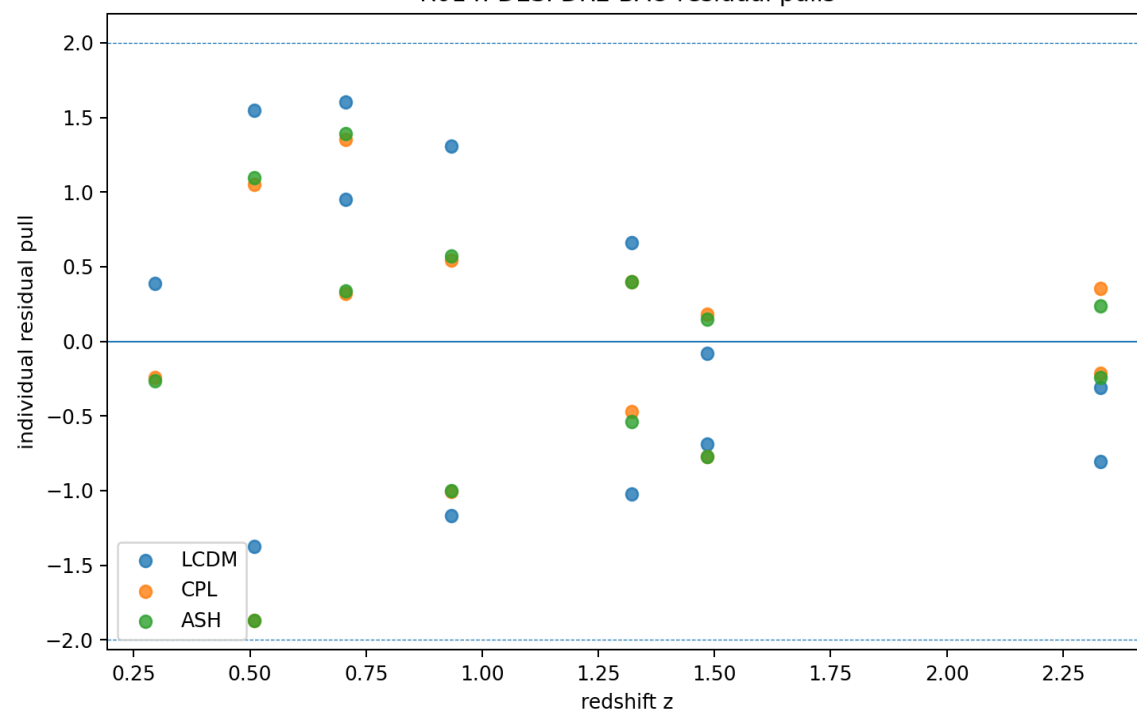
R012: ASH finite-spectral background residual



R012: derived ASH equation of state

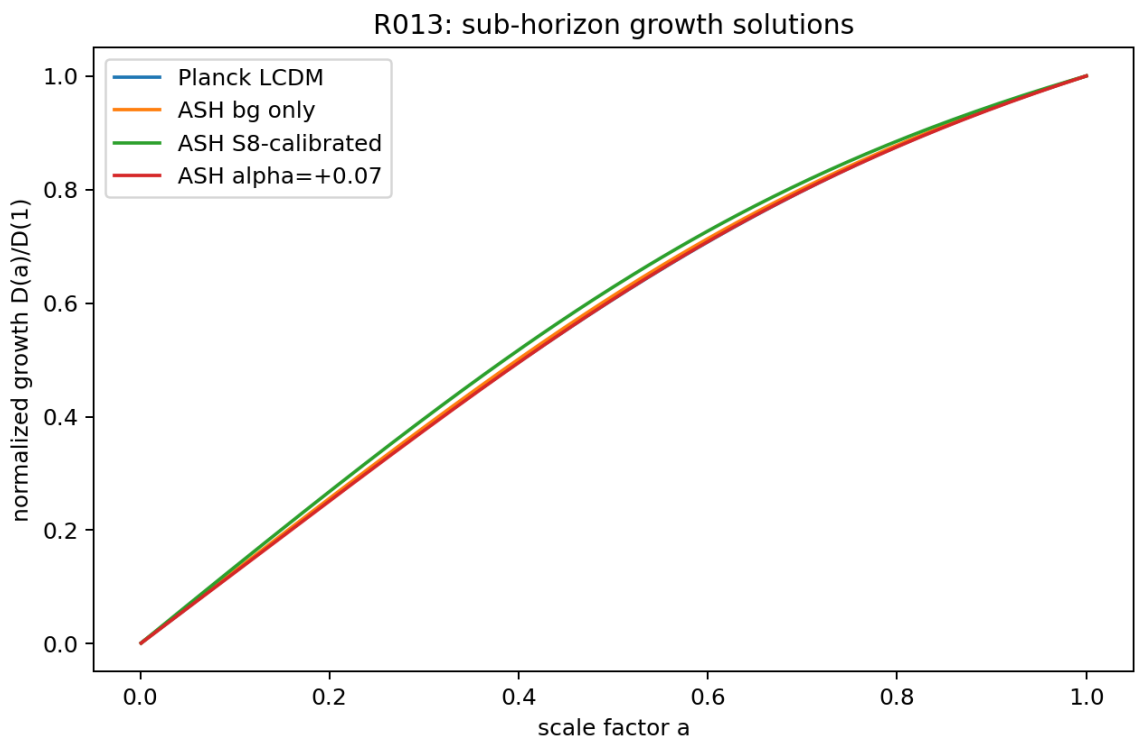
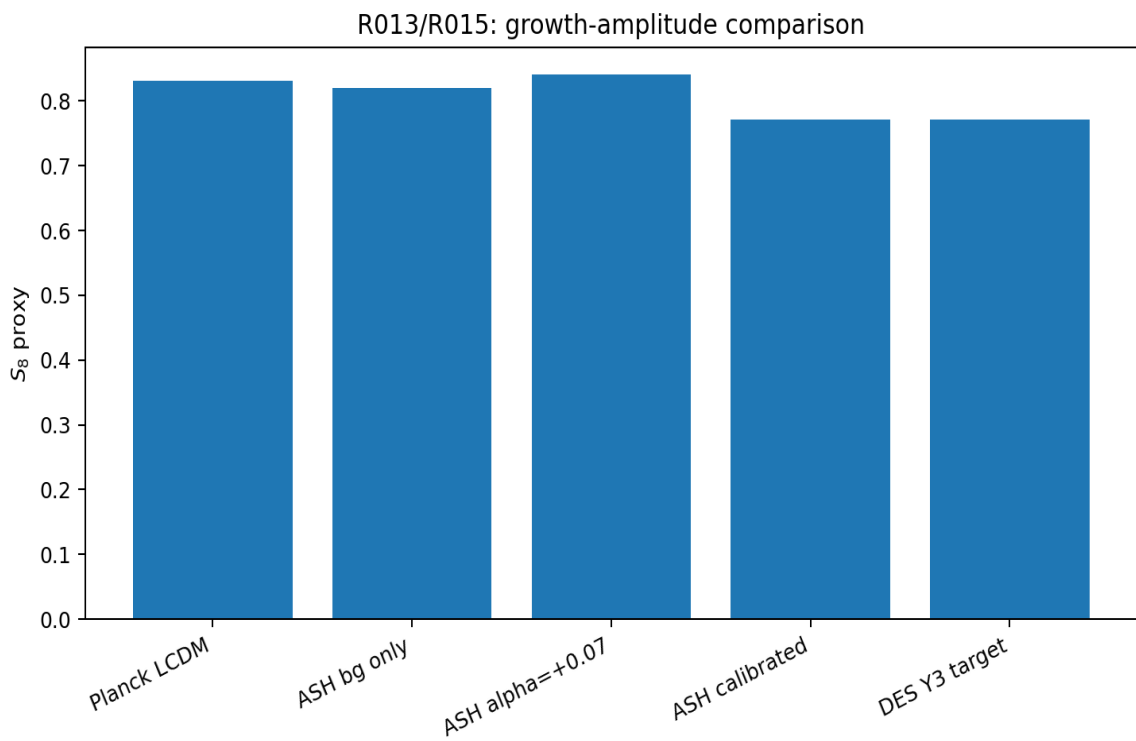


R014: DESI DR2 BAO residual pulls



# R013: growth and S8 calibration

The package solves the sub-horizon growth equation  $D'' + [2 + d \ln H / d \ln a] D' - (3/2) \Omega_m(a) \mu(a) D = 0$  with  $\mu(a) = 1 + \alpha_\mu a^{p_1}$ . The ASH background alone gives  $S_8 = 0.820367$ . The locked positive  $\alpha_\mu = +0.07$  gives  $S_8 = 0.840450$ . DES Y3  $S_8 = 0.772$  requires  $\alpha_\mu = -0.174245$ , yielding  $S_8 = 0.772000$ .



# R015: locked prediction pilot scoring

## ASH-R015-P001 -- not\_confirmed\_by\_pilot\_fit

The DESI+Planck calibrated ASH finite-spectral background requires percent-level residuals and develops a small high- $z$  sign crossing, unlike the locked 0.29% nonnegative envelope. This is not a formal preregistered falsification of P001, but it is a substantive failed pilot match.

```
{
  "max_delta_H_over_H": 0.024557382987014584,
  "z_at_max": 0.315,
  "delta_H_over_H_at_z_1": 0.009125546412692342,
  "min_delta_H_over_H": -0.0006004167000576796,
  "z_at_min": 2.5
}
```

## ASH-R015-P002 -- tension\_in\_sign\_and\_magnitude\_under\_S8\_proxy

The locked positive  $\alpha_{\mu}=0.07$  increases the S8 proxy; the DES Y3 compression requires  $\alpha_{\mu}\approx-0.174$  in this finite-spectral growth equation. A full  $P(k)$  covariance likelihood is still required for formal falsification.

```
{
  "alpha_mu_required_for_DESY3_S8": -0.17424534303771316,
  "S8_background_only": 0.820367303251754,
  "S8_locked_alpha_mu_plus_0p07": 0.8404496497286015,
  "S8_calibrated": 0.7719999994127477,
  "DESY3_S8": 0.772,
  "locked_alpha_pull_sigma_vs_DESY3": 4.02644998403538
}
```

## ASH-R015-P003 -- sign\_consistent\_only

The exact finite ASH spectrum gives a positive odd-over-even proxy for all tested relaxation times; published CMB anomaly literature reports odd-over-even preference in temperature. This is only a sign check and not an empirical likelihood.

```
{
  "finite_spectrum_sign": "positive_odd_over_even",
  "tau_matching_locked_A_2_30_proxy": 1.3988903555840269,
  "A_at_tau": 0.025911762310551152,
  "A_at_tau_1": 0.015437392795883364
}
```

# R016: formal closure certificate

This package closes a deterministic first-pass finite-kernel-to-observable tuple  $M_{\text{ASH-SDE}}=(X, K_I, \text{Spec } K_I, \rho_X(a), E(a), D(a), L_{\text{BAO}}, S_{\text{R015}})$ . All results are reproducible from the included script, compact data tables, and fixed source assumptions.

```
{
  "R012_background_equation": "closed for finite-spectral FRW extension in this package",
  "R013_growth_observable": "closed for sub-horizon scalar growth and S8 proxy in this package",
  "R014_real_data_likelihood": "closed for compact DESI DR2 BAO covariance pilot and compressed priors",
  "R015_locked_prediction_execution": "pilot-scored for P001-P003; P001/P002 not confirmed, P003 sign only",
  "R016_formal_closure": "closed as deterministic finite-kernel-to-observable map with explicit limitations"
}
```

## Remaining limitations

This is not a full validation of ASH as observed cosmology. It does not derive Einstein equations from ASH, does not include a full photon-baryon Boltzmann hierarchy, does not run a map-level CMB likelihood, and does not ingest full supernova, weak-lensing, or galaxy-clustering covariance products. These are the next science tasks, not documentation tasks.

## Key package files

scripts/reproduce\_ash\_r012\_r016\_science.py; outputs/model\_fit\_summary.csv/json; outputs/growth\_predictions.csv;  
outputs/locked\_prediction\_pilot\_scores.json/csv; outputs/r016\_formal\_closure\_certificate.json;  
formal/formal\_expressions\_latex.json; figures/\*.png.